

# Updated U.S. population standard for the Veterans RAND 12-item Health Survey (VR-12)

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## Abstract

**Purpose** The purpose of this project was to develop an updated U.S. population standard for the Veterans RAND 12-item Health Survey (VR-12).

**Methods** We used a well-defined and nationally representative sample of the U.S. population from 52,425 responses to the Medical Expenditure Panel Survey (MEPS) collected between 2000 and 2002. We applied modified regression estimates to update the non-proprietary 1990 scoring algorithms. We applied the updated standard to the Medicare Health Outcomes Survey (HOS) to compute the VR-12 physical (PCS<sub>(MEPS standard)</sub>) and mental (MCS<sub>(MEPS standard)</sub>) component summaries based on the MEPS. We compared these scores to PCS and MCS based on the 1990 U.S. population standard.

**Results** Using the updated U.S. population standard, the average VR-12 PCS<sub>(MEPS standard)</sub> and MCS<sub>(MEPS standard)</sub> scores in the Medicare HOS were 39.82 (standard deviation [SD] = 12.2) and 50.08 (SD = 11.4), respectively. For the same Medicare HOS, the average PCS and MCS scores based on the 1990 standard were 1.40 points higher and 0.99 points lower in comparison to VR-12 PCS and MCS, respectively.

**Conclusions** Changes in the U.S. population between 1990 and today make the old standard obsolete for the VR-12, so the updated standard developed here is widely available to serve as such a contemporary standard for future applications for health-related quality of life (HRQoL) assessments.

**Keywords** Health-related quality of life · Veterans RAND 12-item Health Survey · Veterans · Standard-based scoring

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## Introduction

Health-related quality of life (HRQoL) assessments are frequently used in quality improvement activities, health

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care system accountability, public reporting, and evaluating treatment effectiveness in clinical trials [1–3]. The RAND 36-item (a.k.a. SF-36®) and the SF-12® Health Surveys and the similar Veterans RAND 36-item (VR-36) and 12-item (VR-12) Health Surveys are among the most commonly used HRQoL instruments [4, 5]. The interpretation of scores computed from responses to these HRQoL survey instruments has been enhanced with the use of standard-based (sometimes called norm-based) scoring [6]. Standard-based scoring makes it possible to interpret scores by comparing them to those of a reference population [7, 8]. Scores, therefore, are interpreted as departures from the standard.

There has been increased interest in the use of the VR-12, a health survey questionnaire in the public domain. The VR-12 has been previously adopted by two large national health surveys: the Ambulatory Care Survey of Healthcare Experiences of Patients (SHEP) [9], sponsored by the Veterans Health Administration, and the Medicare Health Outcomes Survey (HOS) [10], sponsored by the Centers for Medicare and Medicaid Services. The current scoring algorithm for the VR-12 is based on the scoring algorithm for version 1 of the SF-12®, which incorporates a 1990 non-institutionalized U.S. population standard [11]. A more recent algorithm for scoring the SF-12®, based on a 1998 U.S. population sample, remains proprietary [12]. A non-proprietary algorithm is preferred, as it facilitates greater scientific scrutiny and more widespread adoption. The non-proprietary 1990 standard, however, needs to be updated, since the composition of the U.S. population has changed over the past decade [13, 14].

The objective of this study was to update the non-proprietary 1990 scoring algorithms for scoring the VR-12. To do this, we used the Medical Expenditure Panel Survey (MEPS), a nationally representative sample of the non-institutionalized general U.S. population, collected between 2000 and 2002. First, we examined the measurement properties of the SF-12® in the MEPS data. Then, we applied modified regression estimation (MRE) algorithms to the MEPS SF-12® data as a means to calibrate the VR-12 responses to those of the SF-12® for the update of the 1990 scoring algorithms. Last, we used the updated standard to compute VR-12 physical and mental summary scores in the Medicare HOS and compared them with scores computed using the original scoring rules from 1990.

## Methods

### Instruments

We used several different but related HRQoL survey instruments: SF-36® version 1, SF-12® version 1, and VR-12. For the remainder of this paper, we will use the terms SF-36® and SF-12® to refer to version 1. The SF-36®

measures eight concepts commonly represented in health surveys: physical functioning (PF), social functioning (SF), role limitation due to physical problems (RP), role limitation due to emotional problems (RE), mental health (MH), energy and vitality (VT), bodily pain (BP), and general perception of health (GH) [15]. The eight scales can be summarized into physical (PCS) and mental (MCS) component summaries using a linear weighting of the 1990 scores based on the 1990 National Opinion Research Center (NORC) norms [11, 16]. These are referred to as the scoring algorithm 1990. The SF-12® is a 12-item subset of the SF-36® that was intended to yield comparable PCS and MCS scores [17, 18].

The VR-12 is a health questionnaire developed from the VR-36, which was modified from the SF-36® [19]. The VR-36 and the VR-12 differ from the SF-36® and SF-12® in the use of five-point response choices for the questions that comprise two scales: role limitations due to physical and emotional problems [20]. Changing the yes/no choices to a five-point response scale (“no, none of the time” to “yes, all of the time”) for seven items in the VR-36 and four items for the VR-12 resulted in a reduction of floor and ceiling effects, with important gains in the scales’ distributional properties and increases in reliability and validity. Validated conversion formulas have been developed for comparisons of VR-36 and VR-12 scores to the SF-36® and the SF-12® Health Surveys [21].

### Data sources

We used three data sources, which included:

1. *Medical Expenditure Panel Survey (MEPS)* This study provided SF-12® data for the development of a more contemporary U.S. population standard [22]. MEPS is co-sponsored by the Agency for Healthcare Research and Quality (AHRQ) and the National Center for Health Statistics (NCHS). The sampling frame for the MEPS is drawn from a nationally representative subset of respondents to the prior year’s National Health Interview Survey (NHIS), conducted by the NCHS [23]. In 2000 and 2001, MEPS incorporated the oversampling of African-Americans and Hispanics that was part of the NHIS sampling design. In 2002, in addition to African-Americans and Hispanics, Asians and others predicted to be poor (<200% of the poverty income level) were also oversampled. The full-year response rates for the 3 years were 65.8% for 2000, 66.3% for 2001, and 64.7% for 2002. Between 2000 and 2002, MEPS collected SF-12® data by providing a self-administered questionnaire (SAQ) to eligible adults (age ≥ 18 years old). MEPS interviewers distributed hard copies of the SAQ to members of sampled households; completed SAQs

were returned by mail. Of the 45,808 adults who received the SAQ, 40,788 answered the questionnaire. Since SF-12<sup>®</sup> data could be inaccurate if reported by a proxy, we eliminated from the analysis 3,845 individuals because someone other than the intended recipient completed the questionnaire. The SAQ was administered once each year, and some respondents completed the SAQ twice during the two-year observation period. We kept both records for the analysis and adjusted for clustering using standard survey variance estimators. The analytic sample included 52,425 SF-12<sup>®</sup> observations from 36,943 respondents.

2. *The Medicare HOS* This study provided SF-36<sup>®</sup> and VR-12 data. We used SF-36<sup>®</sup> data from baseline cohorts 1 (1998) and 2 (1999) to apply the MRE algorithm, described below. We used VR-12 data from cohort 7 follow-up (2006) and cohort 9 baseline (2006) to compute the physical and mental summary scores using the updated scoring algorithm. The Medicare HOS administers mail surveys with telephone follow-ups of non-respondents based on a random sample of 1,000 beneficiaries from each health plan with a Medicare Advantage Program contract. Since 1998, a new cohort is enrolled each year with 2 years of follow-up. Details about Medicare HOS data collection are described elsewhere [24]. The response rates for cohorts 1 and 2 were 61.4 and 66.5%, respectively. The response rate for cohort 7 follow-up was 83.0% (the baseline cohort 7 response rate was 65.3%) and for cohort 9 baseline it was 66.8%.
3. *The Employee Health Care Value Survey (EHCVS)* This study provided SF-36<sup>®</sup> data to conduct a sensitivity analysis of the MRE approach to developing the scoring algorithm. This 154-item health and satisfaction survey questionnaire was administered by mail in 1993 and 1995 to 14,587 subjects. They were randomly selected active employees at six geographic sites in 1993 who served as a longitudinal panel for data collection in 1993 and 1995. Details about the data collection from the EHCVS are described elsewhere [25].

#### The development of the updated population standard

In view of the differences among the data sources and among the versions of the health survey questionnaires (SF-36<sup>®</sup>, SF-12<sup>®</sup>, and the VR-12), we needed to address a number of methodological issues in the process of developing an updated standard. One consideration was the imputation of missing data. A second issue was differences in responses due to context effects. Context effects reflect influences of questionnaire formatting and adjacent questions. The 12 survey questions for the analysis of the

“SF-12” may be drawn from the longer SF-36<sup>®</sup> (a “non-native” 12-items) or may be administered as a reduced questionnaire like the SF-12<sup>®</sup> or the VR-12 (a “native” 12-items) [26]. For example, for the item describing limitations in climbing stairs (pf04), the “native” mail version tended to produce a value 2.59 points higher than the non-native mail version, on average. For these reasons, the SF-12 scoring algorithm was not sufficient to score survey populations using the SF-12 items from the MEPS.

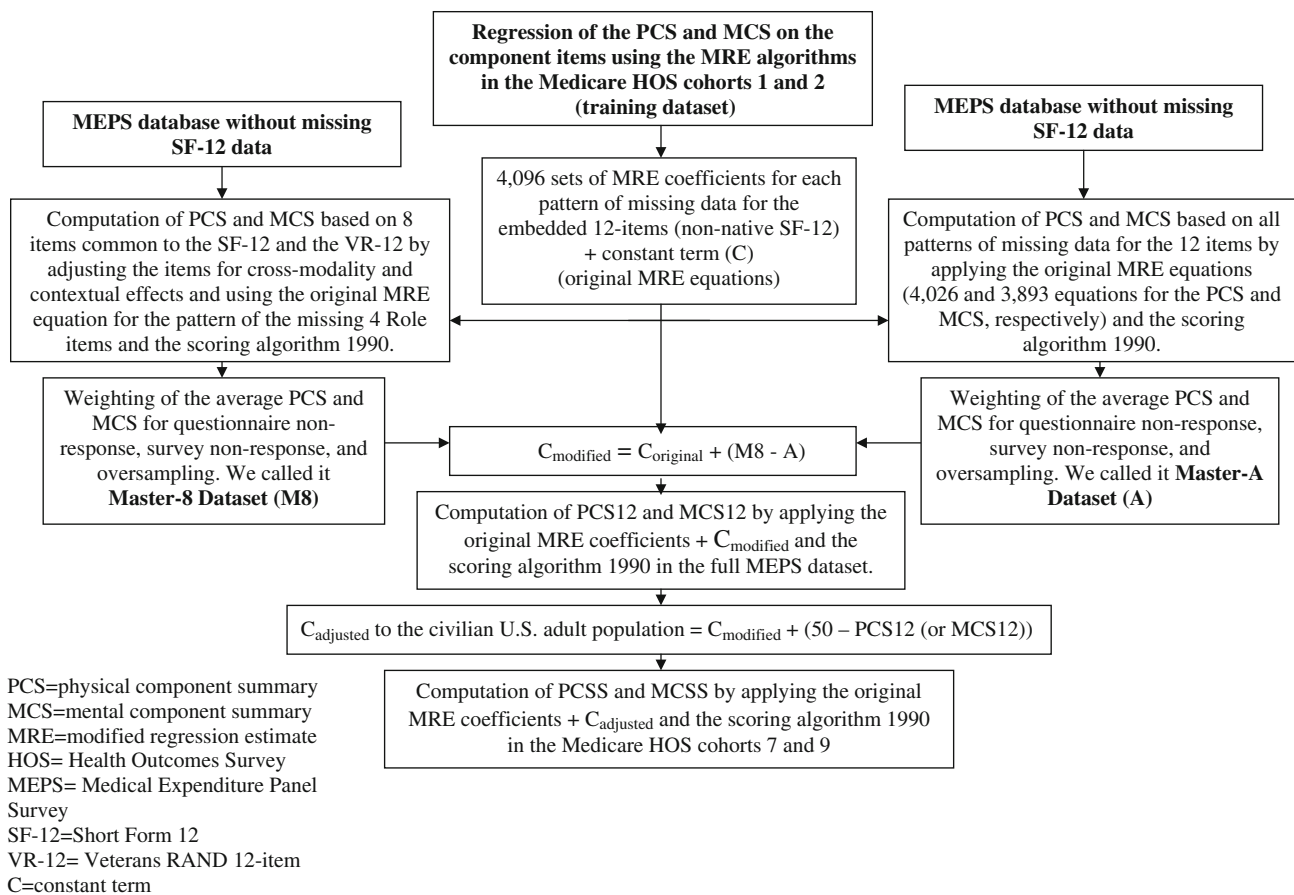
The MRE provides a consistent framework for addressing these two issues [27]. It is a method for calculating a summary score (e.g., PCS-12 or MCS-12) given some or all of the constituent items [28]. As the name implies, it is obtained by applying linear regression in a training dataset where the fully calculated summary score is known. Although the original SF-12 scoring algorithm employed non-linear scoring for each item, assigning a different coefficient to each response option, subsequent research indicates support for linear associations between item scores and the underlying health domains [12]. Thus, we considered only linear (equal interval) coding for each SF-12 item.

Operationally (Fig. 1), we regressed the PCS and MCS on the SF-12 items embedded in the SF-36<sup>®</sup> (non-native SF-12), using data from the Medicare HOS baseline cohorts 1 and 2 (i.e., the training dataset,  $n = 326,482$ ). We used cases with complete data to simulate each potential missing data pattern. We estimated a different regression model for each pattern of missing data for the 12 items, resulting in 4,095 sets of coefficients (after eliminating the situation in which all 12 items had missing data). Each set of these “original MRE coefficients” consisted of one coefficient for each non-missing item, plus a constant. We used those equations with  $R^2$  greater than 0.5 for the analysis, which resulted in the elimination of 68 and 203 sets of equations for PCS and MCS, respectively. The resulting equations were assembled into a dataset of regression coefficients, one for PCS and another for MCS, with a row for each regression equation and 13 columns (one for each SF-12 item and the regression constant).

The MRE is very efficient to calculate using the following algorithm:

1. Label the constituent items 1 to  $k$ .
2. Assign to each item a binary digit, 0 or 1, according to whether that item is missing.
3. Array the binary digits in order and compute the numeric equivalent. This is called a key.
4. Merge the item dataset with a coefficient dataset on the key.
5. Compute the product of the component items and the coefficients, to obtain a predicted score.

The MRE is obtained by taking the regression prediction ( $P$ ) and modifying it with the following formula:



**Fig. 1** The development of the modified constant to update the U.S. population standard

$$MRE = \frac{P - M}{R} + M$$

where  $M$  is the mean of the summary score in the training sample,  $R$  is the multiple correlation coefficient (the square root of the  $R^2$ ), and  $P$  is the score estimated from the regression equation. Thus, a respondent missing items 1 and 7 would be matched with coefficients from a regression model predicting PCS (or MCS) from 10 of the 12 SF-12 items, excluding items one and seven. The coefficients of this model would be used in conjunction with the data from this respondent to predict the dependent variable ( $P$ ), and the predicted value for this person would then be adjusted using the formula above.

Next, we applied the MRE equations derived above to the MEPS data. To be consistent with the previous analysis, we used only cases with complete SF-12<sup>®</sup> data ( $n = 48,585$ ) and simulated missing data patterns (4,026 and 3,893 equations for the PCS and MCS, respectively). We computed the PCS and MCS scores for each pattern of missing data. We used non-adjusted items and the original MRE equations with the scoring algorithm 1990. We averaged them across the whole sample using the analytic weights,

which adjust for questionnaire non-response, survey non-response, and oversampling. We called it “master A.”

Subsequently, we developed adjustments to deal with two issues: (1) deriving the MRE using “non-native” SF-12 data and applying the algorithm to “native” SF-12 data; and (2) the difference between the SF-12 and the VR-12 in response scales for four items (RP and RE scales). To derive the contextual effects, we compared the item changes among baseline and follow-up cohorts of the Medicare HOS that used the VR-12 with previous baseline and follow-up cohorts that received the SF-36<sup>®</sup>. We found that the distributions of responses to “native” items were similar to the distributions for the corresponding embedded (“non-native”) items, but the means (on a 0–100 scale) differed slightly. We concluded that the shift was constant for each item, implying that the regression coefficients used in the MRE are equivalent for both “native” and “non-native” items, and that only the mean for each item needed to be adjusted. Thus, we used the correction factors from Table 1 for the eight items that overlap between the VR-12 and the SF-12<sup>®</sup>.

Next, in order to compute PCS and MCS scores based on the eight items common to the SF-12<sup>®</sup> and the VR12,



we adjusted the items for contextual effects using the correction factors in Table 1 and applied the original MRE equation for the pattern in which the four RP and RE items were missing<sup>1</sup> with the scoring algorithm 1990. We used observations in MEPS with complete SF-12<sup>®</sup> data ( $n = 48,585$ ). We averaged the predicted PCS (MCS) summary scores over the sample, using the analytic weights. We called this average the “master 8.”

We then adjusted the “original” constant term (“ $C_{\text{original}}$ ”) in the dataset of original MRE coefficients (derived from the HOS training data) by the difference between the “master 8” and the “master A” summary scores. This constant adjustment was applied to all keys. The new coefficient dataset had modified constant terms (“ $C_{\text{modified}}$ ”), but the other regression coefficients in each row remained unchanged. We then calculated PCS (and MCS) scores for each MEPS respondent, including those with missing data, using the MRE coefficients with the modified constant terms.

Finally, we adjusted the PCS and MCS scores so that they would have means of 50 in the general 2000–2002 civilian U.S. population. Using the full MEPS sample, we added the overall difference between the average PCS score in the full MEPS sample and 50 to the modified constant terms, resulting in adjusted constant terms (“ $C_{\text{adjusted}}$ ”). We performed a similar adjustment for MCS scores. To facilitate ease of conversion, we did not adjust for differences in the standard deviation (SD); in these data, the SD is close to 10 for both VR-12 PCS<sub>(MEPS standard)</sub> and MCS<sub>(MEPS standard)</sub>. To distinguish the resulting estimated new scores based on the MEPS standard from the older PCS and MCS scores that use different standards, we named the scores VR-12 PCS<sub>(MEPS standard)</sub> and MCS<sub>(MEPS standard)</sub>.

### Statistical analyses

The analyses proceeded in three steps. First, we conducted factor analyses to examine the measurement properties of the MEPS SF-12<sup>®</sup> data. Confirmatory factor analyses were conducted for those observations with complete SF-12<sup>®</sup> data ( $n = 48,585$ ). Four items were recoded, such that higher scores represented better health for all items. Because the response categories were on ordinal response scales, analyses were based on the matrix of polychoric correlations. Weighted least squares estimation was conducted using Mplus software (version 4.1). Analyses also incorporated the analytic weights developed for the MEPS self-administered questionnaire and adjusted standard errors for the complex sampling design.

<sup>1</sup> We applied the following formula:  $(GH1 - 0.84) * \beta_{gh1} + (PF02 - 1.64) * \beta_{pf02} + (PF04 - 2.59) * \beta_{pf04} + (BP2 + 1.27) * \beta_{bp2} + (VT2 - 5.83) * \beta_{vt2} + (MH3 - 6.01) * \beta_{mh3} + (MH4 + 2.77) * \beta_{mh4} + (SF2 - 0.50) * \beta_{sf2} + C$ .

**Table 1** Correction terms for contextual effects

| Item <sup>a</sup>            |       |
|------------------------------|-------|
| General health (gh1)         | −0.84 |
| Moderate activities (pf02)   | −1.64 |
| Climb several flights (pf04) | −2.59 |
| Pain interference (bp2)      | +1.27 |
| Had a lot of pep (vt2)       | −5.83 |
| Felt calm (mh3)              | −6.01 |
| Felt blue (mh4)              | +2.77 |
| Social activities (sf2)      | −0.50 |

<sup>a</sup> The role physical and emotional factors were not included

Second, we used VR-12 data from the Medicare HOS follow-up cohort 7 (2006) and baseline cohort 9 (2006) to compare the VR-12 PCS<sub>(MEPS standard)</sub> and MCS<sub>(MEPS standard)</sub> scores based on the MEPS standard with the PCS and MCS scores based on the 1990 general U.S. population standard. Using the Medicare HOS data, we presented the means, medians (50th percentile), 25th and 75th percentiles, SDs, and observed ranges of the VR-12 PCS<sub>(MEPS standard)</sub> and MCS<sub>(MEPS standard)</sub> scores, standardized to means of 50 in the general population, for the overall population, as well as subgroups based on age, gender, and selected conditions (diabetes, asthma/COPD, high blood pressure, arthritis, heart-related conditions, stroke, and depression).

Third, as a sensitivity analysis, we examined the robustness of the MRE approach in a separate healthy employee data set, the EHCVS. This analysis was performed because the MRE algorithm was developed using the Medicare HOS database, which uses a sample of respondents 65 years of age or older. In contrast, the EHCVS sample comprises younger employed adults. Using the EHCVS as training data for the MRE could result in different regression results and, therefore, a different scoring algorithm. Therefore, we compared these scores with those obtained from the new MEPS-based standard scoring using the HOS data.

## Results

### Sample characteristics

Analyses were conducted on 52,425 observations from 36,943 MEPS respondents. The study participants had a mean age of 45 years ( $SD = 17$ ), with 45% males, 73% whites, 54% married, and 19% with less than high school education.

### Confirmatory factor analysis (CFA)

Based on prior research [29, 30], we specified a model in which five items (moderate activities, climbing stairs, pain,

and the two role-physical items) loaded on a physical health factor, four items (feeling calm, not feeling depressed, and the two role-emotional items) loaded on a mental health factor, and the remaining three items (general health, vitality, and social activities) had loadings on both factors. All loadings were statistically significant (Appendix Table 5). The correlation between the factors was 0.585.

CFA models are typically evaluated in terms of goodness-of-fit indices, which summarize the discrepancies between the observed covariances and the covariances predicted by the model [31, 32]. The comparative fit index (CFI) and the Tucker–Lewis index (TLI) compare the substantive model to a null model of complete independence; values over 0.95 (with a maximum of 1.0) are proposed as indicating acceptable fit. The root mean square error of approximation (RMSEA) measures badness-of-fit per degree of freedom; values lower than 0.05 have been proposed as indicating good fit, and values lower than 0.08 as indicating acceptable fit. In the MEPS data, the CFI was 0.946 and the TLI was 0.979. The RMSEA was 0.087. Overall, the results of CFA analyses suggested adequate fit, although with some room for improvement. The inspection of residuals suggested the presence of secondary associations, especially between items that came from the same SF-36 subscale, such as moderate activities and climbing stairs, or items from the role limitations due to physical problems and role limitations due to emotional problems scales. Freeing the restriction that these residual associations are zero would improve the model's fit, but at the expense of added complexity and difficulty of interpretation. Additional analyses (results not shown) compared subgroups defined by gender, age, and the presence or absence of a chronic condition. Overall, the results were similar for all subgroups.

#### MEPS standards

After applying the modified constant terms in the MRE algorithm for the full MEPS sample, the average SF-12®

PCS and MCS scores were 48.604 (SD = 11.1) and 51.008 (SD = 10.0), respectively. To standardize the scores to the U.S. population (mean 50), we added 1.396 (50–48.604) to the constant term in PCS and subtracted 1.008 (50–51.008) from the constant term in MCS to compute the VR-12 PCS<sub>(MEPS standard)</sub> and MCS<sub>(MEPS standard)</sub>, respectively.

The mean PCS and MCS calculated for the MEPS data using the 1990 standard were 49.39 and 51.15, respectively. Thus, for 51,532 observations with data for the PCS, MCS, VR-12 PCS<sub>(MEPS standard)</sub>, and MCS<sub>(MEPS standard)</sub>, the mean difference between the VR-12 PCS<sub>(MEPS standard)</sub> and PCS was –0.61, and the mean difference between the MCS<sub>(MEPS standard)</sub> and MCS was 1.13.

Table 2 shows that the average VR-12 PCS<sub>(MEPS standard)</sub> and MCS<sub>(MEPS standard)</sub> scores in the Medicare HOS (follow-up cohort 7 and baseline cohort 9) were 39.81 (±12.2) and 50.08 (±11.4), respectively. The average VR-12 PCS<sub>(MEPS standard)</sub> and MCS<sub>(MEPS standard)</sub> scores were 1.40 points higher and 0.99 points lower, respectively, compared with the average PCS and MCS scores based on the non-institutionalized general U.S. population 1990 standard.

Table 3 shows the VR-12 PCS<sub>(MEPS standard)</sub> and MCS<sub>(MEPS standard)</sub> mean, median (50th percentile), 25th and 75th percentiles, SD, and observed range (minimum and maximum) of scores for the overall population and subgroups based on age, gender, and co-morbidities. The age group 64 years and younger, who were eligible for Medicare based on a disability, had lower VR-12 PCS<sub>(MEPS standard)</sub> and MCS<sub>(MEPS standard)</sub> scores than those 65 years and older, who were eligible for Medicare primarily based on age. Among the disability groups as well as the older Americans, the VR-12 PCS<sub>(MEPS standard)</sub> and MCS<sub>(MEPS standard)</sub> scores decreased with age. Males had a slightly higher VR-12 PCS<sub>(MEPS standard)</sub> and MCS<sub>(MEPS standard)</sub> scores than females. Standardized data for individuals with various chronic conditions showed that respondents with more physically morbid conditions, such as stroke, had lower scores on the VR-12 PCS<sub>(MEPS standard)</sub>, on average, than

**Table 2** Comparisons between the average VR-12 physical (PCS) and mental (MCS) health summary scores based on the Medical Expenditure Panel Survey (MEPS) standard and the average physical

(PCS) and mental (MCS) component summary scores based on the general U.S. 1990 standard in the Medicare Health Outcome Survey (HOS) follow-up cohort 7 and baseline cohort 9

|                    | PCS <sub>(MEPS standard)</sub> | PCS <sub>(1990 standard)</sub> | MCS <sub>(MEPS standard)</sub> | MCS <sub>(1990 standard)</sub> |
|--------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|
| Mean               | 39.82                          | 38.42                          | 50.08                          | 51.09                          |
| 25th percentile    | 30.52                          | 29.12                          | 42.90                          | 43.90                          |
| 50th percentile    | 40.71                          | 39.31                          | 53.14                          | 54.14                          |
| 75th percentile    | 50.41                          | 49.01                          | 58.94                          | 59.94                          |
| Standard deviation | 12.29                          | 12.28                          | 11.49                          | 11.49                          |
| Minimum            | 0.587                          | –0.809                         | –2.473                         | –1.465                         |
| Maximum            | 72.11                          | 70.71                          | 76.09                          | 77.09                          |
| <i>n</i>           | 173,221                        | 173,221                        | 172,432                        | 172,432                        |

PCS = physical component summary; MCS = mental component summary; higher PCS and MCS scores indicate better health

**Table 3** VR-12 physical (PCS) and mental (MCS) health summary scores based on the Medical Expenditure Panel Survey (MEPS) standard for the overall population and subgroups based on age, gender, and selected conditions in the Medicare Health Outcome Survey (HOS) follow-up cohort 7 (2006) and baseline cohort 9 (2006)

|                | PCS <sub>(MEPS standard)</sub> |       |       |       |       |         |         | MCS <sub>(MEPS standard)</sub> |       |       |       |       |       |         |         |         |
|----------------|--------------------------------|-------|-------|-------|-------|---------|---------|--------------------------------|-------|-------|-------|-------|-------|---------|---------|---------|
|                | Mean                           | SD    | 75th  | 50th  | 25th  | Maximum | Minimum | n                              | Mean  | SD    | 75th  | 50th  | 25th  | Maximum | Minimum | n       |
| Overall sample | 39.82                          | 12.29 | 50.41 | 40.71 | 30.52 | 72.11   | 0.59    | 173,221                        | 50.08 | 11.49 | 58.94 | 53.14 | 42.90 | 76.09   | −2.47   | 172,432 |
| Subgroups      |                                |       |       |       |       |         |         |                                |       |       |       |       |       |         |         |         |
| Age 18–34      | 40.76                          | 12.98 | 50.82 | 41.92 | 32.44 | 63.97   | 5.90    | 184                            | 41.67 | 14.31 | 53.42 | 41.46 | 31.31 | 70.21   | 6.41    | 181     |
| Age 35–44      | 34.01                          | 12.86 | 43.51 | 32.85 | 23.61 | 62.99   | 2.40    | 790                            | 37.58 | 14.61 | 49.78 | 37.05 | 26.88 | 68.37   | 2.70    | 784     |
| Age 45–54      | 29.48                          | 10.76 | 36.33 | 27.65 | 21.63 | 68.04   | 3.89    | 2,332                          | 36.03 | 13.58 | 45.28 | 35.71 | 26.04 | 76.09   | −1.90   | 2,316   |
| Age 55–64      | 28.72                          | 10.35 | 34.97 | 27.19 | 21.38 | 66.31   | 1.90    | 5,733                          | 39.86 | 13.17 | 49.31 | 39.44 | 30.74 | 73.41   | 2.48    | 5,711   |
| Age 65–74      | 43.04                          | 11.71 | 53.01 | 45.11 | 35.00 | 72.11   | 1.18    | 55,036                         | 51.85 | 10.35 | 59.51 | 55.27 | 45.97 | 74.81   | −2.47   | 54,827  |
| Age 75+        | 38.43                          | 12.01 | 48.26 | 39.06 | 29.36 | 68.06   | 0.59    | 56,615                         | 49.92 | 11.38 | 58.92 | 52.41 | 42.61 | 75.97   | 3.76    | 56,221  |
| Males          | 40.67                          | 12.19 | 51.29 | 41.91 | 31.68 | 68.44   | 0.59    | 70,794                         | 50.37 | 11.43 | 59.16 | 53.73 | 43.35 | 76.09   | 0.46    | 70,483  |
| Females        | 39.22                          | 12.32 | 49.68 | 39.95 | 29.79 | 72.11   | 0.82    | 102,427                        | 49.88 | 11.53 | 58.91 | 52.78 | 42.60 | 75.97   | −2.47   | 101,949 |
| Diabetes       | 35.96                          | 12.02 | 45.21 | 36.02 | 26.48 | 67.61   | 0.59    | 38,160                         | 47.91 | 12.28 | 58.30 | 50.07 | 39.45 | 74.28   | 4.28    | 38,153  |
| Hypertension   | 38.33                          | 12.12 | 48.37 | 38.88 | 29.02 | 70.88   | 0.59    | 108,441                        | 49.55 | 11.64 | 58.78 | 52.33 | 41.96 | 76.09   | 0.46    | 108,405 |
| Asthma         | 33.36                          | 11.71 | 41.86 | 32.72 | 24.19 | 67.22   | 0.59    | 24,828                         | 46.75 | 12.87 | 57.59 | 48.44 | 37.68 | 76.09   | 4.91    | 24,818  |
| Heart disease  | 35.53                          | 11.97 | 44.71 | 35.47 | 26.06 | 70.88   | 0.70    | 58,684                         | 48.11 | 12.29 | 58.31 | 50.42 | 39.64 | 75.97   | −1.90   | 58,666  |
| Stroke         | 32.85                          | 11.46 | 40.98 | 32.15 | 23.91 | 66.81   | 0.70    | 15,552                         | 45.32 | 13.08 | 56.40 | 46.50 | 36.12 | 76.09   | 4.95    | 15,540  |
| Arthritis      | 36.25                          | 11.78 | 45.29 | 36.36 | 27.01 | 68.44   | 0.70    | 9,2511                         | 48.67 | 12.02 | 58.43 | 50.98 | 40.53 | 76.09   | 2.37    | 9,2486  |
| Depression     | 34.61                          | 11.63 | 42.86 | 33.94 | 25.32 | 72.11   | 1.18    | 50,546                         | 39.89 | 11.73 | 48.21 | 39.87 | 32.23 | 74.06   | −2.47   | 50,540  |
| No conditions  | 49.69                          | 8.85  | 56.21 | 52.72 | 45.46 | 71.35   | 5.33    | 1,6579                         | 55.78 | 6.81  | 60.50 | 58.19 | 52.69 | 72.96   | 1.84    | 1,6573  |

PCS = physical component summary; MCS = mental component summary; higher PCS and MCS scores indicate better health; SD = standard deviation; n = sample size

**Table 4** Average SF-12<sup>®</sup> PCS and MCS scores based on the Medical Expenditure Panel Survey (MEPS) and the Employee Health Care Value Survey (EHCVS) modified constant terms (sensitivity analysis) in the MEPS full sample

| MEPS modified constant terms SF-12 <sup>®</sup> in the MEPS full sample |       |      | EHCVS modified constant terms SF-12 <sup>®</sup> in the MEPS full sample |       |      |
|-------------------------------------------------------------------------|-------|------|--------------------------------------------------------------------------|-------|------|
| <i>n</i>                                                                | Mean  | SD   | <i>n</i>                                                                 | Mean  | SD   |
| PCS 52,226                                                              | 48.60 | 11.1 | 52,198                                                                   | 48.63 | 11.2 |
| MCS 52,203                                                              | 51.01 | 10.1 | 52,192                                                                   | 50.27 | 10.3 |

PCS = physical component summary; MCS = mental component summary; higher PCS and MCS scores indicate better health; SD = standard deviation; *n* = sample size

those respondents reporting less physically disabling conditions, such as hypertension. Those who reported depression had lower scores on the MCS<sub>(MEPS standard)</sub>, on average, than those respondents reporting medical conditions.

### Sensitivity analysis

We repeated Steps 1–4 in the development of the scoring standard, using the EHCVS sample as the training database to develop the regression models. The MRE coefficients based on the EHCVS data and the HOS data were similar (Appendix Table 6). The mean PCS and MCS scores at the end of Step 4, calculated using the MEPS data, were 48.63 for the PCS and 50.27 for the MCS (Table 4). These means were very close to the means of 48.60 and 51.01 obtained when the MRE algorithm used training data from HOS cohorts 1 and 2. Although the HOS data were from those 65 years of age or older or disabled persons and the EHCVS data were from healthy working-age adults, the similarity in the resulting standards strongly suggests the robustness of the MRE algorithm.

### Discussion

Computing scores relative to a population standard has enhanced the interpretation of HRQoL surveys. Our results indicate that the updated U.S. population standard is useful and robust for calculating VR-12 physical and mental health summary scores. The resulting physical and mental summary scores for the VR-12 vary as expected as a function of gender and selected medical and psychiatric conditions.

Our MRE methodology has several strengths. First, the MRE improves significantly in terms of accuracy, missing data tolerance, and computation burden when compared with best-known proprietary technology, such as the missing data estimation (MDE) software [33] for scoring summary measures such as the PCS and MCS [34]. The sensitivity analysis confirmed the robustness of our MRE approach. Second, computations of VR-12 PCS<sub>(MEPS standard)</sub> and MCS<sub>(MEPS standard)</sub> scores are relatively simple to obtain with an updated standard that is standardized to the scores in a contemporary U.S. population. This is computed simply by

adding 1.396 to the constant term in PCS and by subtracting 1.008 from the constant term in MCS. These correction factors can also be easily applied to average SF-12<sup>®</sup> PCS and MCS scores based on the 1990 standard to calculate the average VR-12 PCS<sub>(MEPS standard)</sub> and MCS<sub>(MEPS standard)</sub> scores. From a software viewpoint, the user does not carry out any item adjustments on his or her own. The new coefficient table does all the work; the user should use unmodified items, scaled 0 to 100. Third, our methodology is not proprietary and in the public domain. Therefore, it can be widely used and be subjected to rigorous scientific review.

The updated standard for the VR-12 is grounded on a well-defined and nationally representative sample of the U.S. population from 52,425 responses to the MEPS. This is important, as concerns were raised with the 1990 standard regarding its small sample size and the possibility of important sample bias. The difference between the MEPS-based standard and the 1990 standard is about 10% of a standard deviation. This is not a trivial difference on a population-wide basis. If we consider prospective mortality studies using PCS and MCS, the relative risk of dying is increased by close to 8% for each point decrease in PCS and 4% for each point decrease in MCS [9, 35].

Prior studies that conducted factor analyses of the SF-12<sup>®</sup> items have obtained results similar to those in our study, consistently obtaining a two-factor solution, with one factor representing physical health status and the second representing mental health status [36–38]. Our overall conclusion is that the fit of the two-factor model is adequate without further model modification and that the psychometric properties of the SF-12<sup>®</sup> items in the MEPS data support summarizing the items by physical and mental health scores.

There are a number of limitations in this study that might affect the results. First, the MRE algorithm was developed using the Medicare HOS database, which uses a mixed sample of young disabled individuals (age 18–64) and older Americans (65 years of age or older). However, we obtained comparable standards using EHCVS data from healthy working-age adults in the sensitivity analysis, indicating the robustness of the MRE algorithm. Second, we were precluded from developing new scoring algorithms for the VR-12 de novo due to the fact that the MEPS used the SF-12<sup>®</sup>, which differs from the VR-12 in the



response format to the four RP and RE items. We were unwilling to assume that factor loadings (or scoring coefficients) derived from an analysis of the SF-12 would apply directly to the VR-12 due to the differences in these four items. Similarly, we could have examined the factor pattern for the eight items that overlap between the SF-12 and the VR-12, but the extent to which removing the four RP and RE items would affect the generalizability of the factor pattern is unclear, and we would still be left with the issue of determining how to score the omitted items.

In summation, we have presented a robust method for deriving physical and mental summary scores for the VR-12 based upon an updated standard from the U.S. population. The standard has a contemporary mean of 50 in the general U.S. population, as represented in the MEPS 2000–2002. The updated U.S. population standard for the VR-12 is publicly accessible (<http://www.va.gov/chqoer/>). The MRE algorithm is a valuable procedure for future HRQoL assessments based on several different but related HRQoL survey instruments and offers the potential for exploring methodological issues related to these health survey questionnaires (e.g., correlated summary scores).

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## Appendix

**Table 5** Confirmatory factor analysis of SF-12 items

| Item                             | Unstandardized     |                    | Standardized       |                    |
|----------------------------------|--------------------|--------------------|--------------------|--------------------|
|                                  | Factor 1           | Factor 2           | Factor 1           | Factor 2           |
| General health                   | 0.646              | 0.257              | 0.590              | 0.181              |
| Moderate activities              | 1.000 <sup>b</sup> | 0.000 <sup>a</sup> | 0.913              | 0.000 <sup>a</sup> |
| Climbing stairs                  | 0.969              | 0.000 <sup>a</sup> | 0.885              | 0.000 <sup>a</sup> |
| Accomplished less (physical)     | 1.047              | 0.000 <sup>a</sup> | 0.956              | 0.000 <sup>a</sup> |
| Limited in activities (physical) | 1.068              | 0.000 <sup>a</sup> | 0.975              | 0.000 <sup>a</sup> |
| Accomplished less (mental)       | 0.000 <sup>a</sup> | 1.329              | 0.000 <sup>a</sup> | 0.939              |
| Limited in activities (mental)   | 0.000 <sup>a</sup> | 1.251              | 0.000 <sup>a</sup> | 0.887              |
| Pain                             | 0.912              | 0.000 <sup>a</sup> | 0.833              | 0.000 <sup>a</sup> |
| Felt calm                        | 0.000 <sup>a</sup> | 1.026              | 0.000 <sup>a</sup> | 0.725              |
| Had a lot of pep                 | 0.363              | 0.735              | 0.332              | 0.519              |
| Felt blue                        | 0.000 <sup>a</sup> | 1.000 <sup>b</sup> | 0.000 <sup>a</sup> | 0.706              |
| Social activities                | 0.325              | 0.844              | 0.297              | 0.596              |

<sup>a</sup> Coefficient constrained to equal zero

<sup>b</sup> Coefficient constrained to 1.0 to set scale of factor

**Table 6** Modified regression estimation (MRE) coefficients (for the cases of no missing values) based on the Employee Health Care Value Survey (EHCVS) data and the Medicare Health Outcomes Survey (HOS) data

| Item                             | Database | Dependent variable |          |
|----------------------------------|----------|--------------------|----------|
|                                  |          | PCS                | MCS      |
| General health                   | EHCVS    | 0.08363            | −0.00046 |
|                                  | HOS      | 0.08575            | 0.00848  |
| Moderate activities              | EHCVS    | 0.07804            | −0.03946 |
|                                  | HOS      | 0.06594            | −0.03149 |
| Climbing stairs                  | EHCVS    | 0.07033            | −0.03287 |
|                                  | HOS      | 0.06929            | −0.03378 |
| Accomplished less (physical)     | EHCVS    | 0.05127            | −0.01224 |
|                                  | HOS      | 0.04639            | −0.01557 |
| Limited in activities (physical) | EHCVS    | 0.05226            | −0.01829 |
|                                  | HOS      | 0.05220            | −0.01530 |
| Accomplished less (mental)       | EHCVS    | −0.03295           | 0.07023  |
|                                  | HOS      | −0.03222           | 0.07078  |
| Limited in activities (mental)   | EHCVS    | −0.02821           | 0.06410  |
|                                  | HOS      | −0.02182           | 0.06400  |
| Pain                             | EHCVS    | 0.14635            | −0.03254 |
|                                  | HOS      | 0.13670            | −0.02514 |
| Felt calm                        | EHCVS    | −0.03440           | 0.10024  |
|                                  | HOS      | −0.03739           | 0.09422  |
| Had a lot of pep                 | EHCVS    | 0.02688            | 0.07555  |
|                                  | HOS      | 0.03266            | 0.05987  |
| Felt blue                        | EHCVS    | −0.04832           | 0.14120  |
|                                  | HOS      | −0.04761           | 0.14389  |
| Social activities                | EHCVS    | −0.00424           | 0.12576  |
|                                  | HOS      | 0.01153            | 0.11122  |
| C-constant                       | EHCVS    | 21.46034           | 16.29832 |
|                                  | HOS      | 21.39888           | 17.57390 |

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